



УНИВЕРСИТЕТ ИТМО

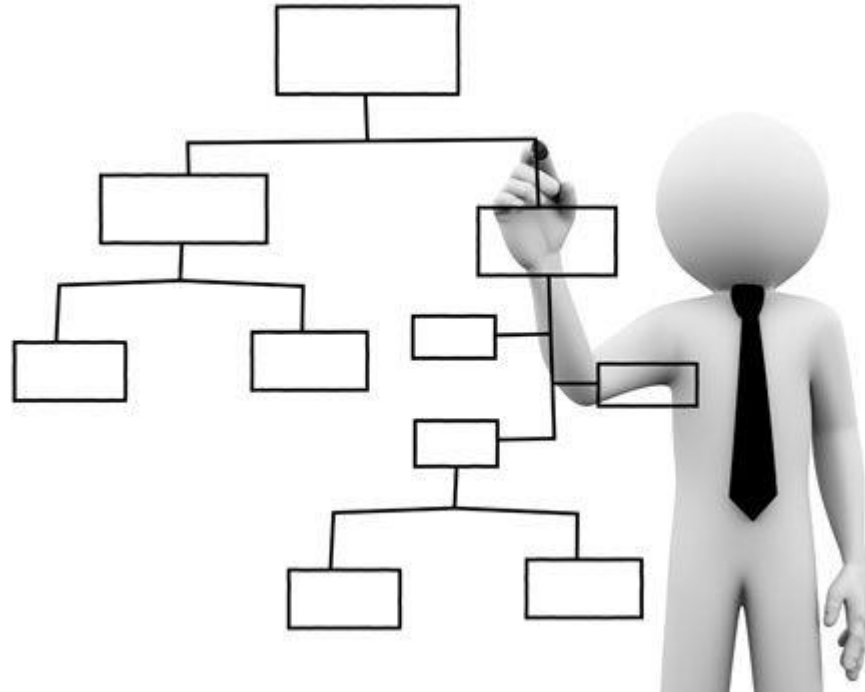
Факультет систем управления и робототехники

Nonlinear systems

Zimenko Konstantin

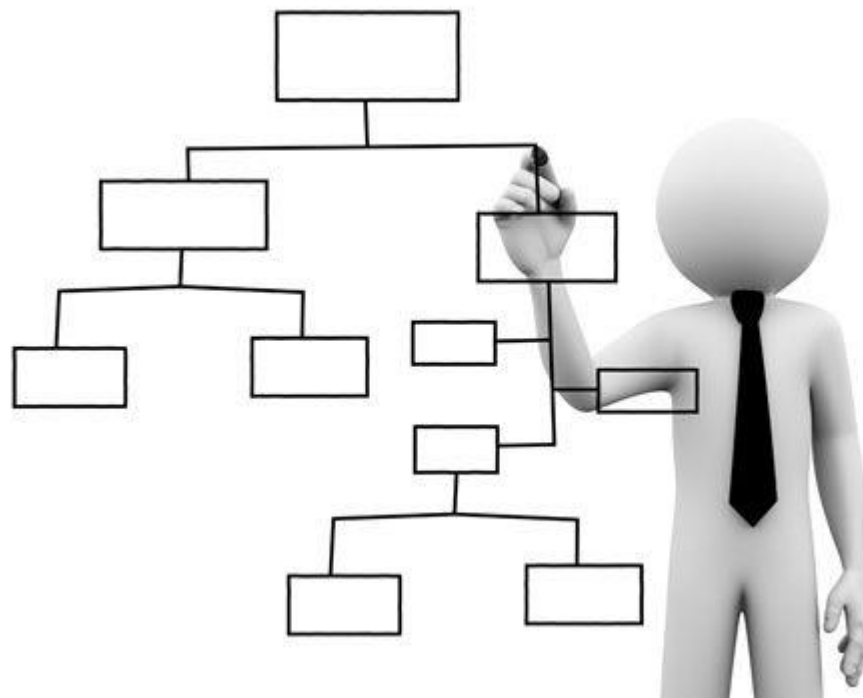
Academic discipline

- 8 lectures
- 8 practical classes

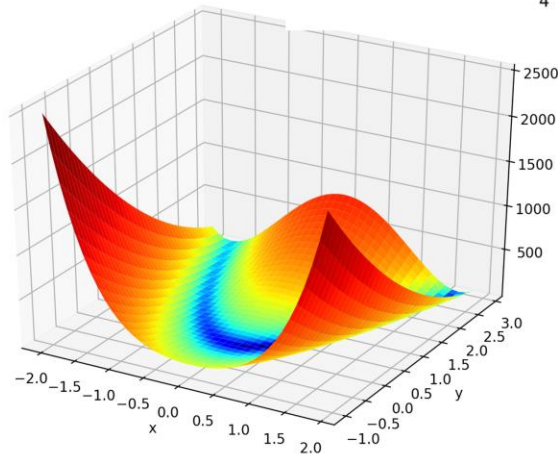
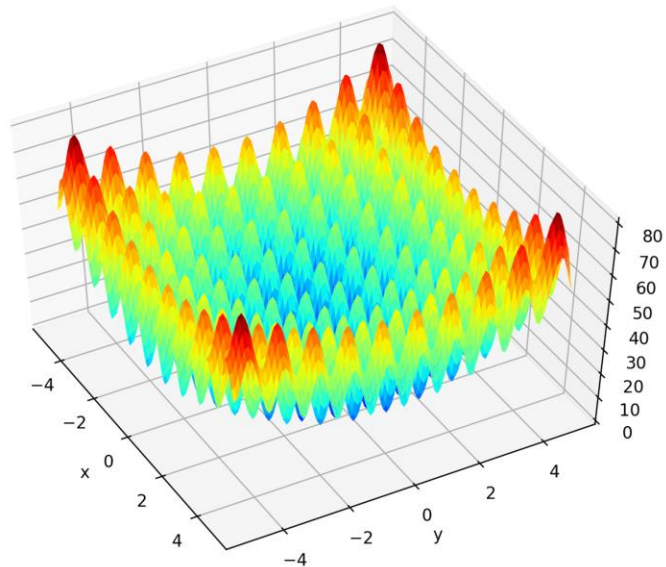
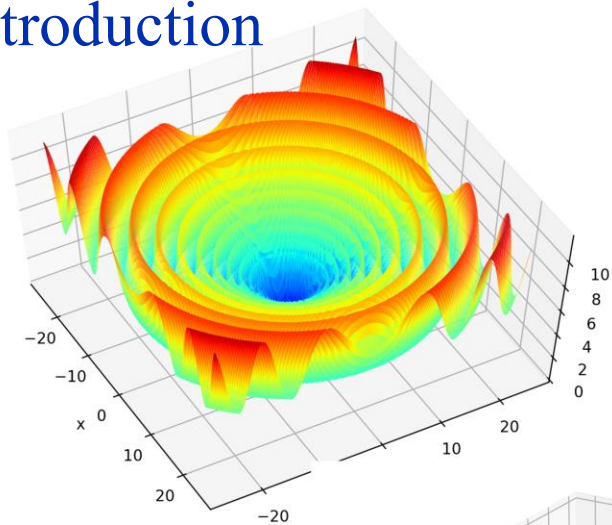


Structure

1. Introduction
2. Equilibrium points and phase portraits
3. Lyapunov function
4. Feedback stabilization
5. Feedback linearization
6. Backstepping
7. Sliding-mode control
- *. LMI



1. Introduction



1. Introduction

- ✔ What is control system, linear system and nonlinear system ?

Control system
is a set of control devices and an object

Linear system
is a system whose mathematical model contains only linear expressions.

Nonlinear system
is a system whose mathematical model contains at least one nonlinear expression.

1. Introduction

✓ Where is the nonlinear system

$$\begin{aligned} 1. \quad & \dot{x}_1 = x_2 \\ & \dot{x}_2 = -5 \sin x_1 - 5x_2 \end{aligned}$$

$$\begin{aligned} 2. \quad & \dot{x}_1 = x_1 + x_2 \\ & \dot{x}_2 = |-9|x_1 - 5^7 x_2 \end{aligned}$$

$$\begin{aligned} 3. \quad & \dot{x}_1 = x_1 x_2 + 2x_2 \\ & \dot{x}_2 = \text{sign}(x_1) - x_2 \end{aligned}$$

$$\begin{aligned} 4. \quad & \dot{x}_1 = x_1 + |2x_2| \\ & \dot{x}_2 = -x_2 \end{aligned}$$

$$\begin{aligned} 5. \quad & \dot{x}_1 = x_1 \\ & \dot{x}_2 = -x_2 \end{aligned}$$

$$\begin{aligned} 6. \quad & \dot{x}_1 = x_1^{0.1} \\ & \dot{x}_2 = -x_2 \end{aligned}$$

$$\begin{aligned} 7. \quad & \dot{x}_1 = 4(x_1 - x_2) \\ & \dot{x}_2 = 4^3 - x_2 \end{aligned}$$

1. Introduction

✓ Nonlinear versus linear systems

Linear systems

- Huge body of work in analysis and control of linear systems
- Most models currently available are linear (but most real systems are nonlinear...)



Nonlinear systems

- Dynamics of linear systems are not rich enough to describe many commonly observed phenomena



Nonlinear systems can (sometime) be approximated by linear systems.
Nonlinear systems can (sometime) be “transformed” into linear systems.

1. Introduction

- ✓ We shall deal with dynamical systems which are modeled by a finite number of coupled first-order ordinary differential equations

$$\begin{aligned}\dot{x}_1 &= f_1(t, x_1, \dots, x_n, u_1, \dots, u_p) \\ \dot{x}_2 &= f_2(t, x_1, \dots, x_n, u_1, \dots, u_p) \\ &\dots \\ \dot{x}_n &= f_n(t, x_1, \dots, x_n, u_1, \dots, u_p)\end{aligned}$$

where $x = [x_1, x_2, \dots, x_n]$ is the state variable, $u = [u_1, u_2, \dots, u_n]$ is the input signal. The symbol $\dot{x} = \frac{dx}{dt}$ denote the derivative of x with respect to time t . With $f = [f_1, f_2, \dots, f_n]$

$$\dot{x} = f(t, x, u)$$

Recall that in the linear case, an explicit dependence on the state vector can be distinguished

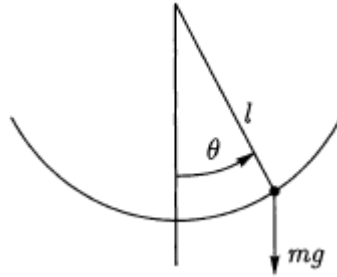
$$\dot{x} = Ax + Bu$$

1. Introduction

✓ Nonlinear systems: Example 1

Pendulum equation (equation of motion in the tangential direction)

$$ml\ddot{\theta} = -mg \sin \theta - kl\dot{\theta}$$



State equations: ???

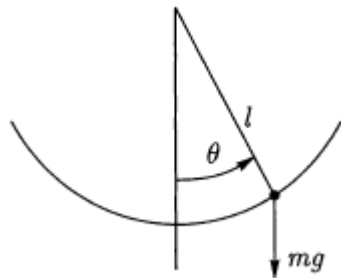
Equilibrium points: ???

1. Introduction

✓ Nonlinear systems: Example 1

Pendulum equation (equation of motion in the tangential direction)

$$ml\ddot{\theta} = -mg \sin \theta - kl\dot{\theta}$$



State equations:

$$(x_1 = \theta, x_2 = \dot{\theta}) \quad \begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{g}{l} \sin x_1 - \frac{k}{m} x_2 \end{aligned}$$

Equilibrium points:

$$(n\pi; 0), n = 0, \pm 1, \pm 2, \dots$$

1. Introduction

✓ Nonlinear systems: Example 1

$$ml\ddot{\theta} = -mg \sin \theta$$

State equations (frictional resistance is neglected): $(x_1 = \theta, x_2 = \dot{\theta})$

Equilibrium points: $(n\pi; 0), n = 0, \pm 1, \pm 2, \dots$

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{g}{l} \sin x_1\end{aligned}$$

✓ Nonlinear systems: Example 2

$$ml\ddot{\theta} = -mg \sin \theta - kl\dot{\theta} + \frac{1}{ml^2} T$$

where T is the torque.

State equations (with friction and applied torque): ???

Equilibrium points: ???

1. Introduction

✓ Nonlinear systems: Example 2

Pendulum equation

$$ml\ddot{\theta} = -mg \sin \theta - kl\dot{\theta} + \frac{1}{ml^2}T$$

State equations (with friction and applied torque):

$$(x_1 = \theta, x_2 = \dot{\theta})$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{g}{l} \sin x_1 - \frac{k}{m}x_2 + \frac{1}{ml^2}T$$

Equilibrium points:

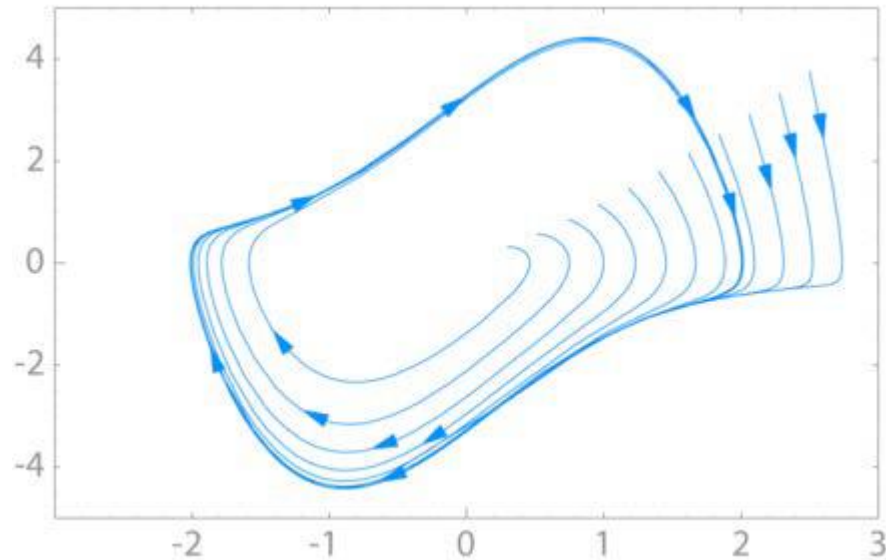
$$\left(\arcsin \left(\frac{T}{mgl} \right); 0 \right)$$

1. Introduction

✓ Nonlinear systems: Example 3

Robust oscillation

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1 + e(1 - x_1^2)x_2\end{aligned}$$

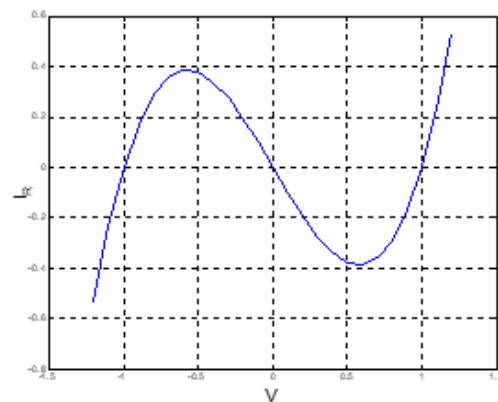
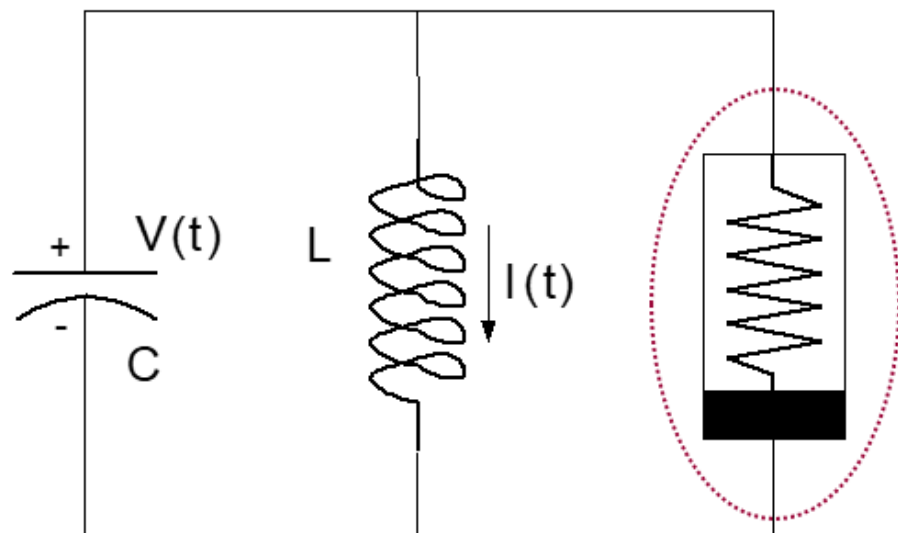


1. Introduction

✓ Nonlinear systems: Example 4

Van der Pol oscillator

$$\begin{aligned}L\dot{I} &= V \\C\dot{V} &= -I - I_R\end{aligned}$$



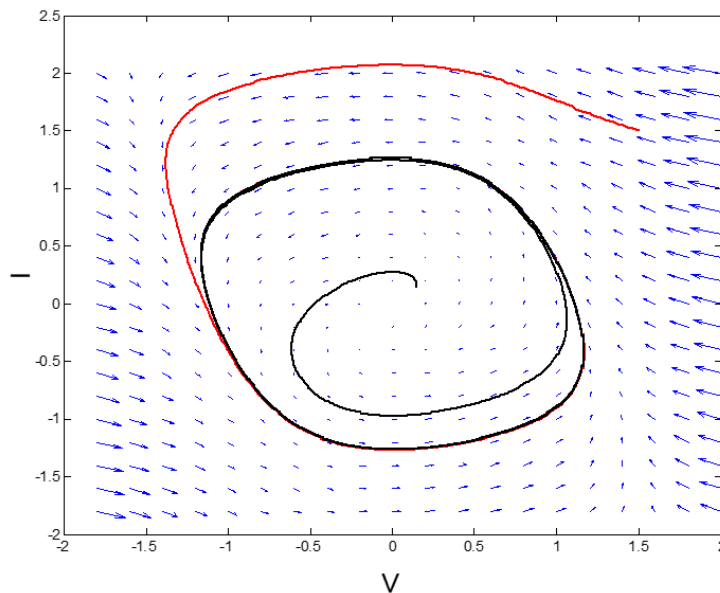
1. Introduction

✓ Nonlinear systems: Example 4

Van der Pol oscillator

$$\begin{aligned}L\dot{I} &= V \\ C\dot{V} &= -I - I_R\end{aligned}$$

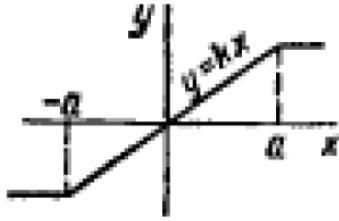
Phase portrait



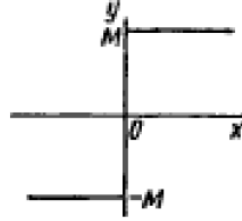
- ✓ Nonlinear phenomena
 - ***Finite escape time:*** (the state of unstable linear system goes to infinity as $t \rightarrow \infty$)
 - ***Nonasymptotic stability (e.g. finite-time stability):*** (linear systems – infinite time of convergence)
 - ***Multiple isolated equilibria:*** (linear systems – only one isolated equilibrium point)
 - ***Limit cycles:*** (linear systems – system oscillates iff there is a pair of eigenvalues on the imaginary axis, which is a nonrobust condition)
 - ***Subharmonic, harmonic, or almost-periodic oscillations:*** (stable linear system under periodic input produces an output of the same frequency)
 - ***Chaos:*** (More complicated steady-state behavior)
 - ***Multiple modes of behavior***

1. Introduction

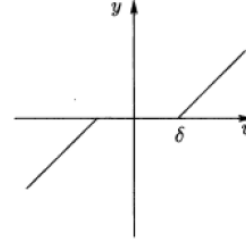
✓ Common nonlinearities



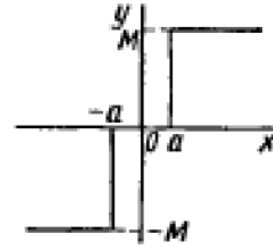
Saturation



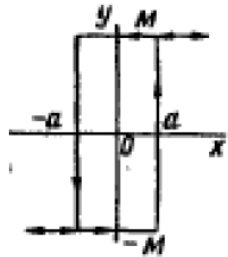
Relay



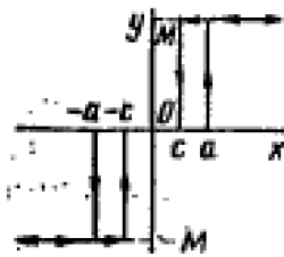
Dead zone



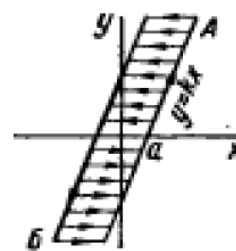
Relay with dead zone



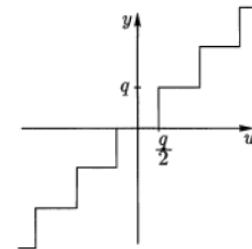
Relay with hysteresis



Three-position relay

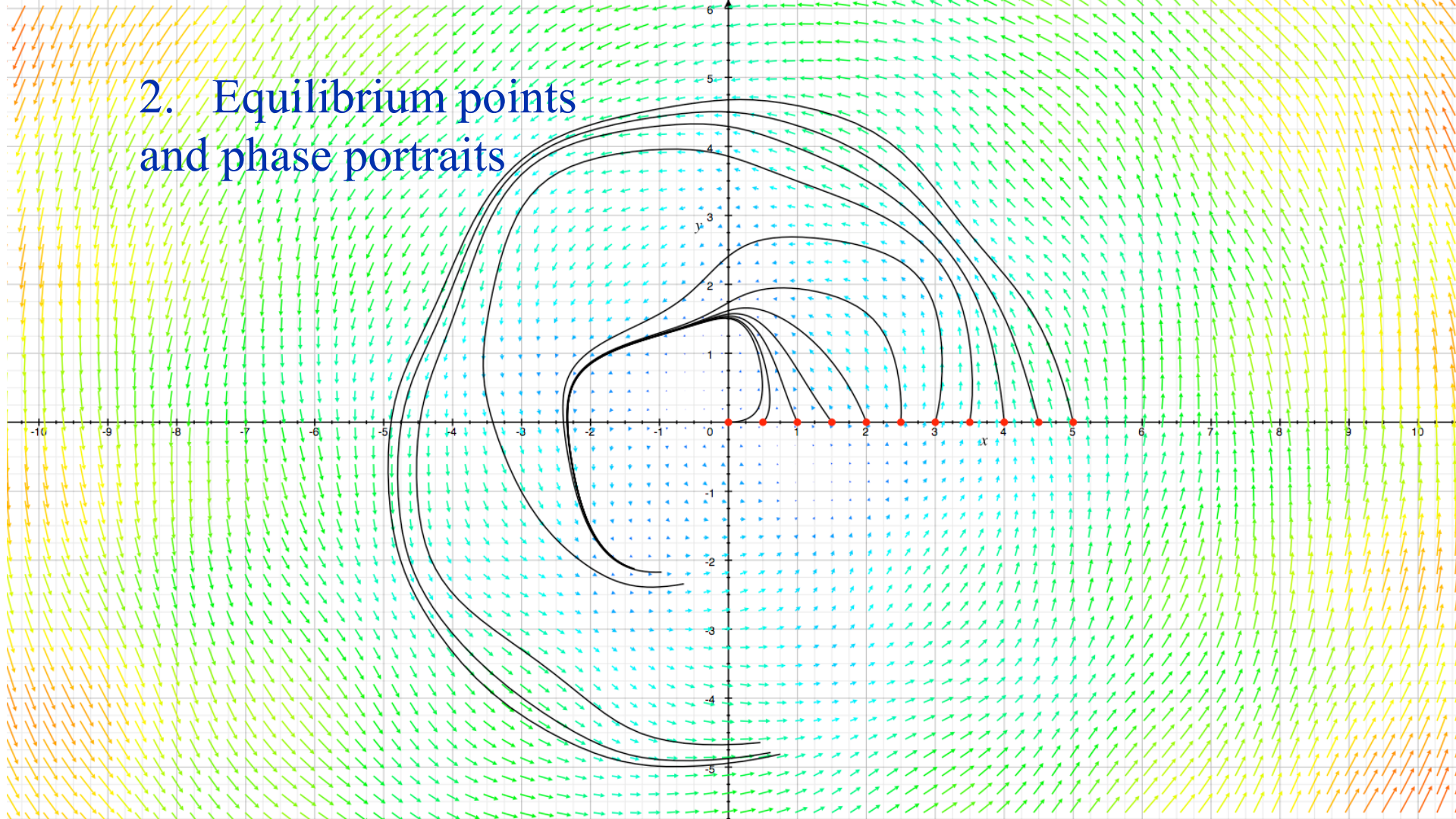


Backlash



Quantization

2. Equilibrium points and phase portraits



2. Equilibrium points and phase portraits

- ✓ Let's remember - Qualitative behavior of linear systems

Linear second order system

$$\dot{x} = Ax, \quad x \in \mathbb{R}^2, \quad A \in \mathbb{R}^{2 \times 2},$$

Apply a similarity transformation M to A :

$$M^{-1}AM = J, \quad M \in \mathbb{R}^{2 \times 2},$$

where J is the real Jordan form of A , which depending on the eigenvalues of A may take one of the three forms

$$\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \quad \begin{bmatrix} \lambda & k \\ 0 & \lambda \end{bmatrix}, \quad \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix},$$

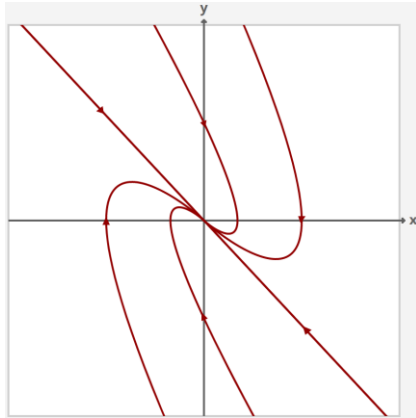
with k being either 0 or 1

Present a change of coordinates

$$\begin{aligned} z &= M^{-1}x, \\ \dot{z} &= M^{-1}\dot{x} \end{aligned}$$

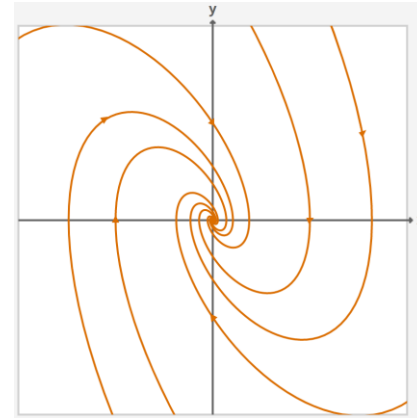
2. Equilibrium points and phase portraits

- ✓ Let's remember - Qualitative behavior of linear systems



Stable node $(\lambda_{1,2} < 0)$

Unstable node $(\lambda_{1,2} > 0)$



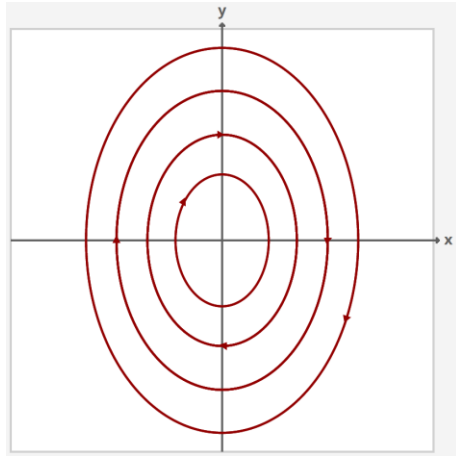
Stable focus $(\lambda_{1,2} = a \pm j\beta)$

Unstable focus $(\lambda_{1,2} = -a \pm j\beta)$

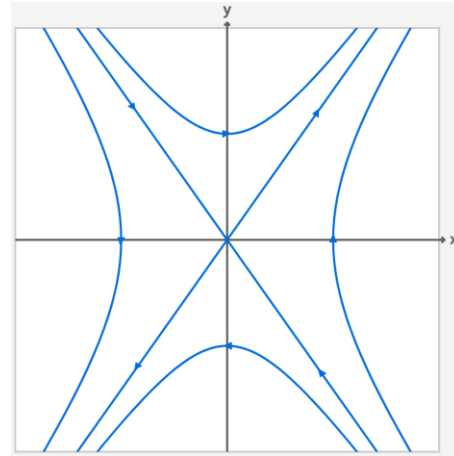
for $a > 0$

2. Equilibrium points and phase portraits

- ✓ Let's remember - Qualitative behavior of linear systems



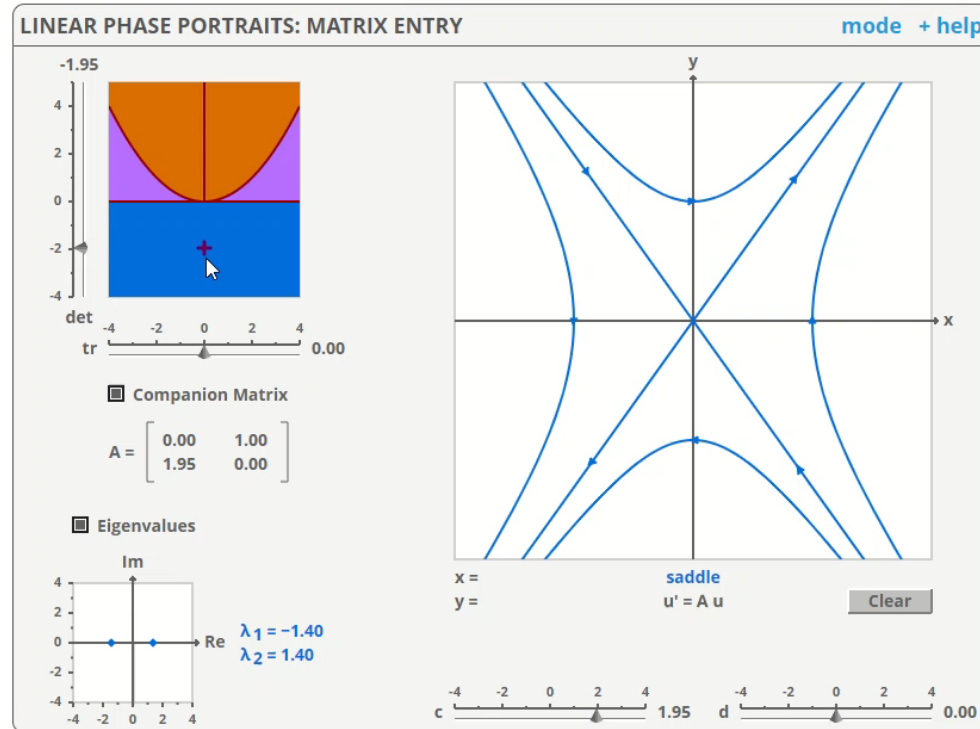
Center $(\lambda_{1,2} = \pm j\beta)$



Saddle point $(\lambda_1 < 0 < \lambda_2)$

2. Equilibrium points and phase portraits

✓ Let's remember - Qualitative behavior of linear systems



<https://mathlets.org/mathlets/linear-phase-portraits-matrix-entry/>

2. Equilibrium points and phase portraits

✓ Behavior of nonlinear systems and Multiple equilibrium points

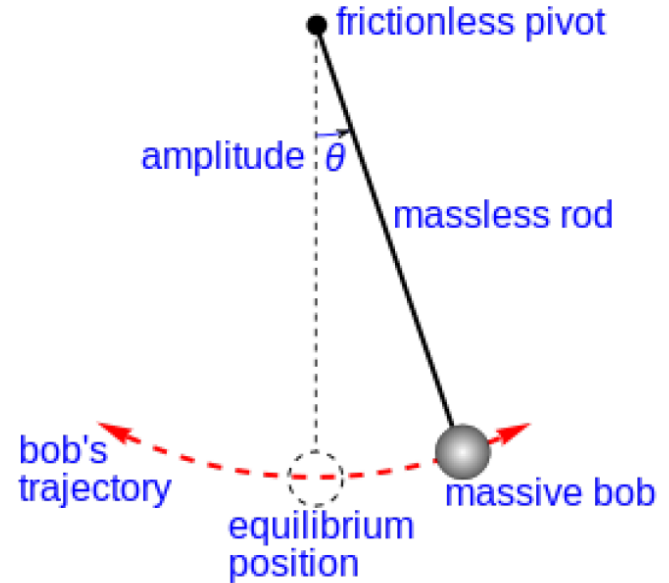
✓ Pendulum

$$\theta = x_1$$

$$\dot{\theta} = x_2$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -10 \sin x_1 - x_2$$



2. Equilibrium points and phase portraits

✓ Behavior of nonlinear systems and Multiple equilibrium points

✓ Pendulum

$$\theta = x_1$$

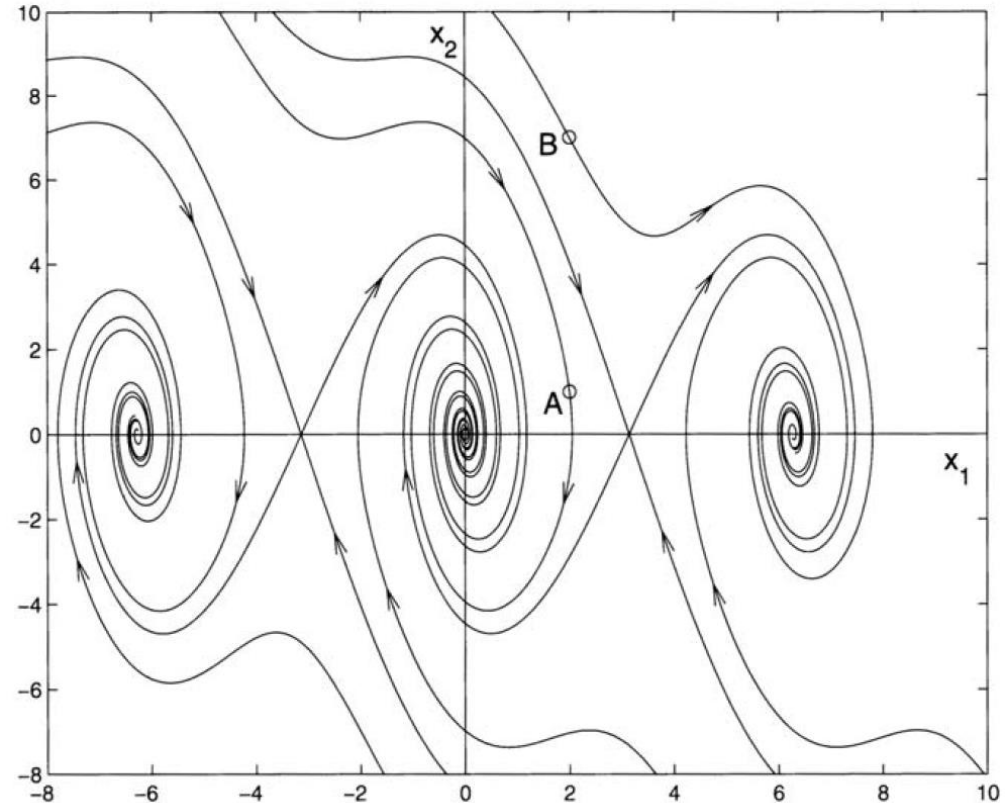
$$\dot{\theta} = x_2$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -10 \sin x_1 - x_2$$

$$A = (c; b_1)$$

$$B = (c; b_2) \quad b_2 > b_1$$



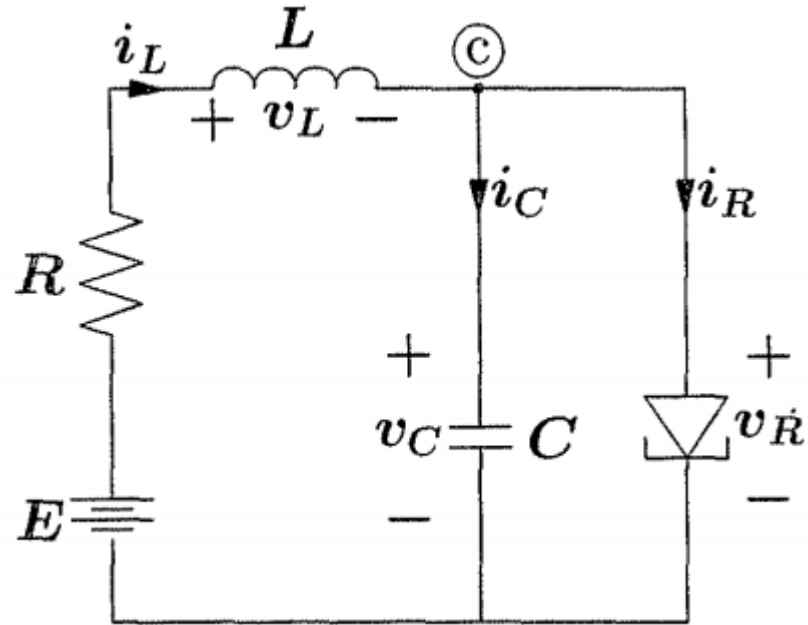
2. Equilibrium points and phase portraits

✓ Behavior of nonlinear systems and Multiple equilibrium points

✓ Tunnel-diode circuit

$$\dot{x}_1 = \frac{1}{C} [-h(x_1) + x_2]$$

$$\dot{x}_2 = \frac{1}{L} [-x_1 - Rx_2 + u]$$



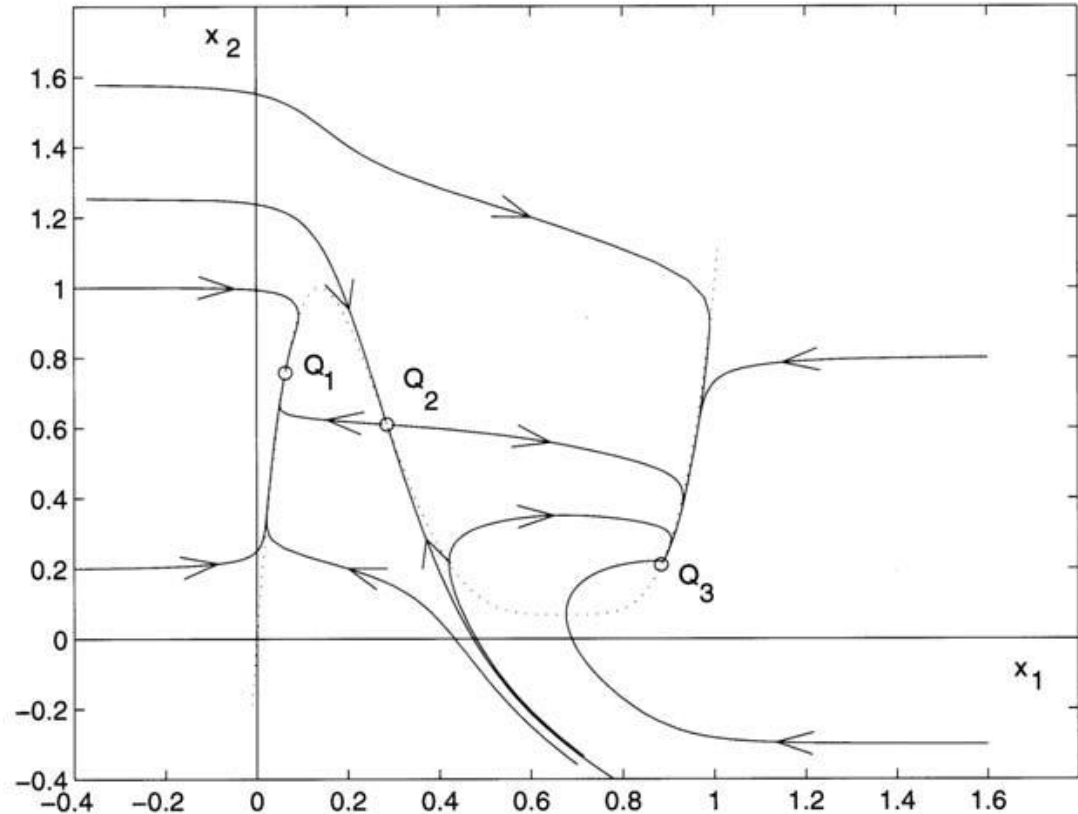
2. Equilibrium points and phase portraits

✓ Behavior of nonlinear systems and Multiple equilibrium points

✓ Tunnel-diode circuit

$$\dot{x}_1 = \frac{1}{C} [-h(x_1) + x_2]$$

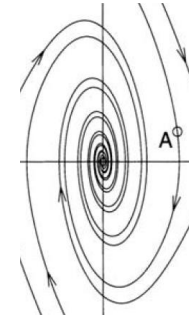
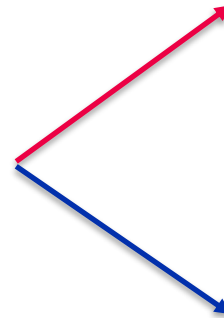
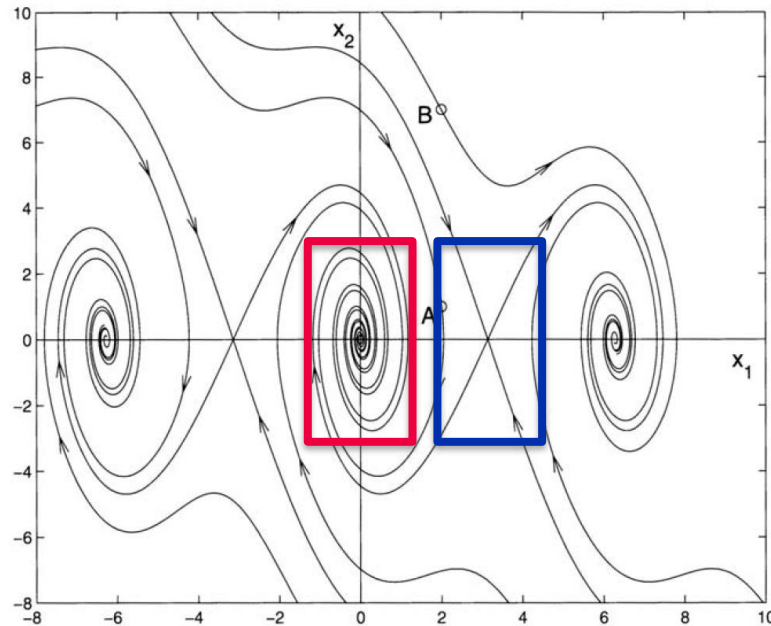
$$\dot{x}_2 = \frac{1}{L} [-x_1 - Rx_2 + u]$$



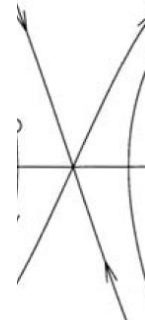
2. Equilibrium points and phase portraits

✓ Note

The qualitative behavior in the vicinity of each equilibrium point looks just like those we saw for linear systems.



???

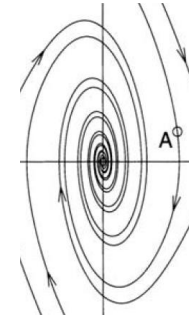
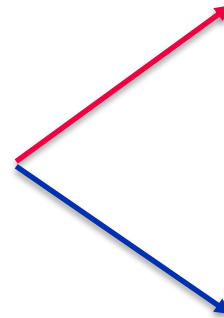
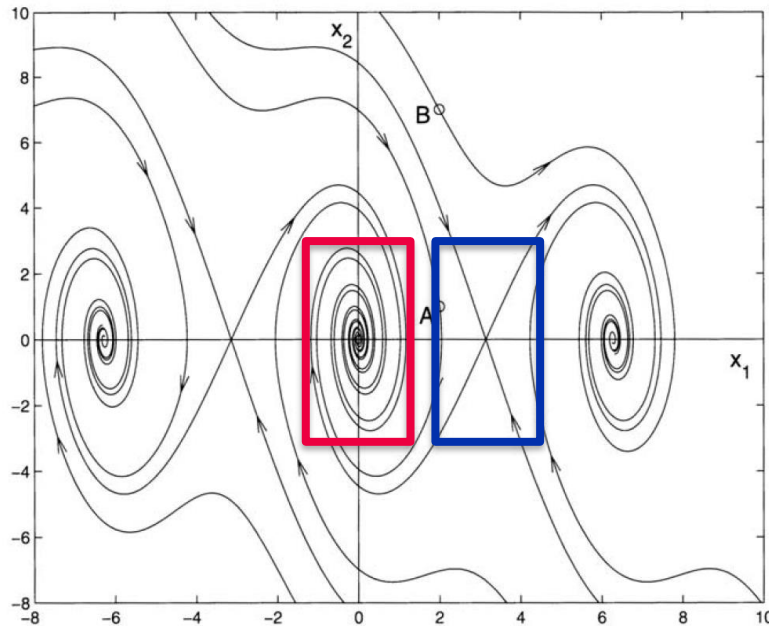


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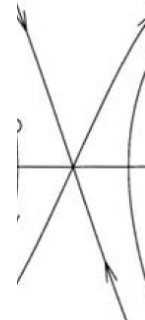
2. Equilibrium points and phase portraits

✓ Note

The qualitative behavior in the vicinity of each equilibrium point looks just like those we saw for linear systems.



Stable focus



Saddle point

2. Equilibrium points and phase portraits

✓ Qualitative behavior near equilibrium

Consider autonomous system

$$\begin{aligned}\dot{x}_1 &= f_1(x_1, x_2) \\ \dot{x}_2 &= f_2(x_1, x_2)\end{aligned}$$

where $f_1(x_1, x_2)$, $f_2(x_1, x_2)$ are continuously differentiable.

Let $p = (p_1, p_2)$ is the **equilibrium point**. Expanding the right-hand side into its Taylor series about the point p , obtain

$$\begin{aligned}\dot{x}_1 &= f_1(p_1, p_2) + a_{11}(x_1 - p_1) + a_{12}(x_2 - p_2) + HOT \\ \dot{x}_2 &= f_2(p_1, p_2) + a_{21}(x_1 - p_1) + a_{22}(x_2 - p_2) + HOT\end{aligned}$$

where *HOT* denotes high order terms and

2. Equilibrium points and phase portraits

$$\begin{aligned}\dot{x}_1 &= f_1(p_1, p_2) + a_{11}(x_1 - p_1) + a_{12}(x_2 - p_2) + HOT \\ \dot{x}_2 &= f_2(p_1, p_2) + a_{21}(x_1 - p_1) + a_{22}(x_2 - p_2) + HOT\end{aligned}$$

where *HOT* denotes high order terms and

$$\begin{aligned}a_{11} &= \left. \frac{\partial f_1(x_1, x_2)}{\partial x_1} \right|_{x_1=p_1, x_2=p_2} & a_{12} &= \left. \frac{\partial f_1(x_1, x_2)}{\partial x_2} \right|_{x_1=p_1, x_2=p_2} \\ a_{21} &= \left. \frac{\partial f_2(x_1, x_2)}{\partial x_1} \right|_{x_1=p_1, x_2=p_2} & a_{22} &= \left. \frac{\partial f_2(x_1, x_2)}{\partial x_2} \right|_{x_1=p_1, x_2=p_2}\end{aligned}$$

Since $p = (p_1, p_2)$ is an equilibrium point

$$f_1(p_1, p_2) = f_2(p_1, p_2) = 0$$

2. Equilibrium points and phase portraits

Define

$$y_1 = x_1 - p_1 \quad \text{and} \quad y_2 = x_2 - p_2$$

and rewrite the state equation as

$$\begin{aligned}\dot{y}_1 &= \dot{x}_1 = a_{11}y_1 + a_{12}y_2 + HOT \\ \dot{y}_2 &= \dot{x}_2 = a_{21}y_1 + a_{22}y_2 + HOT\end{aligned}$$

HOT is negligible in a small neighborhood of equilibrium point:

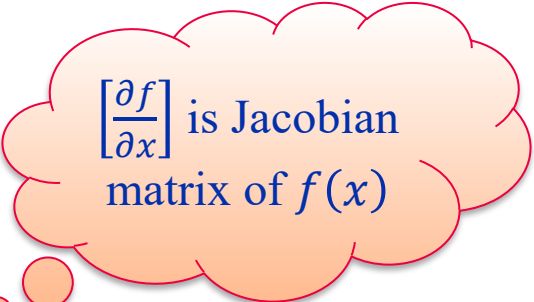
$$\begin{aligned}\dot{y}_1 &= \dot{x}_1 = a_{11}y_1 + a_{12}y_2 \\ \dot{y}_2 &= \dot{x}_2 = a_{21}y_1 + a_{22}y_2\end{aligned}$$

Rewriting in a vector form, obtain

$$\dot{y} = Ay$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \left. \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} \right|_{x=p} = \left. \frac{\partial f}{\partial x} \right|_{x=p}$$



$\left[\frac{\partial f}{\partial x} \right]$ is Jacobian
matrix of $f(x)$

2. Equilibrium points and phase portraits

✓ Nonlinear systems: Example 1

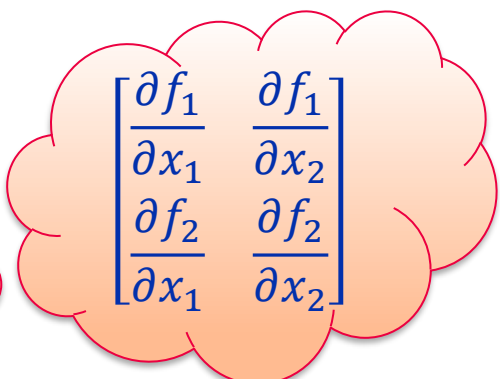
Pendulum equation

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -10 \sin x_1 - x_2\end{aligned}$$

Equilibrium points $(0; 0)$ and $(\pi; 0)$

Jacobian

$$\frac{\partial f}{\partial x} = \begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix}$$


$$\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}$$

2. Equilibrium points and phase portraits

✓ Nonlinear systems: Example 1

Pendulum equation

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -10 \sin x_1 - x_2\end{aligned}$$

Equilibrium points $(0; 0)$ and $(\pi; 0)$

Jacobian

$$\frac{\partial f}{\partial x} = \begin{bmatrix} 0 & 1 \\ -10 \cos x_1 & -1 \end{bmatrix}$$

Jacobian evaluated at the equilibrium point

$$A_1 = \left. \frac{\partial f}{\partial x} \right|_{x=(0;0)} = \begin{bmatrix} 0 & 1 \\ -10 & -1 \end{bmatrix}$$

$$\lambda_{1,2} = -0.5 \pm \frac{3}{12}j$$

$$A_2 = \left. \frac{\partial f}{\partial x} \right|_{x=(\pi;0)} = \begin{bmatrix} 0 & 1 \\ 10 & -1 \end{bmatrix}$$

$$\lambda_1 = -3.7, \quad \lambda_2 = 2.7,$$

2. Equilibrium points and phase portraits

✓ Nonlinear systems: Example 2

Pendulum equation

$$\dot{x}_1 = -x_2 - \mu x_1(x_1^2 + x_2^2)$$

$$\dot{x}_2 = x_1 - \mu x_2(x_1^2 + x_2^2)$$

Jacobian at $(0; 0)$ has eigenvalues $\pm j$.

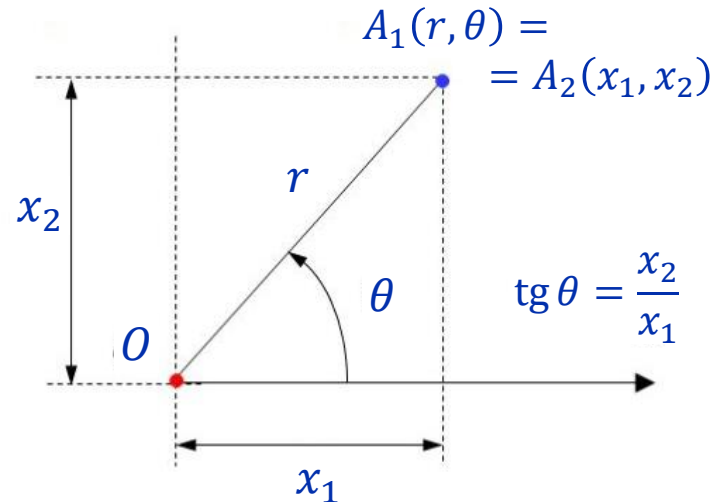
Transition to polar coordinates:

$$x_1 = r \cos \theta$$

$$x_2 = r \sin \theta$$

$$r = \sqrt{x_1^2 + x_2^2}$$

$$\theta = \operatorname{arctg} \frac{x_2}{x_1}$$



2. Equilibrium points and phase portraits

✓ Nonlinear systems: Example 2

Pendulum equation

$$\begin{aligned}\dot{x}_1 &= -x_2 - \mu x_1(x_1^2 + x_2^2) \\ \dot{x}_2 &= x_1 - \mu x_2(x_1^2 + x_2^2)\end{aligned}$$

Jacobian at $(0; 0)$ has eigenvalues $\pm j$.

Transition to polar coordinates:

$$\begin{aligned}x_1 &= r \cos \theta \\ x_2 &= r \sin \theta\end{aligned}$$

$$\begin{aligned}r &= \sqrt{x_1^2 + x_2^2} \\ \theta &= \operatorname{arctg} \frac{x_2}{x_1}\end{aligned}$$

The system in polar coordinates

$$\begin{aligned}\dot{r} &= -\mu r^3 \\ \dot{\theta} &= 1\end{aligned}$$

For $\mu > 0$ the equilibrium point $(0; 0)$ is a **stable focus**, for $\mu < 0$ is an **unstable focus**.

The background is a complex, low-poly geometric pattern in various shades of blue, ranging from dark navy to light sky blue. The shapes are irregular polygons that fit together like a mosaic. In the lower center, there is a white rounded rectangle with a thin white border. Inside this rectangle, the text "IT's MOre than a UNIVERSITY" is written in white. "IT's" is in a small sans-serif font, "MOre" is in a larger sans-serif font with a lowercase 'o', "than a" is in a smaller italicized sans-serif font, and "UNIVERSITY" is in a large, bold, all-caps sans-serif font.

IT's MOre than a
UNIVERSITY